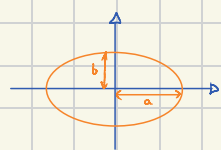


Lecture 18: Change of variables

Changing variables in double integrals

Example 1: area of ellipse with semi-axes a, b



$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = 1$$

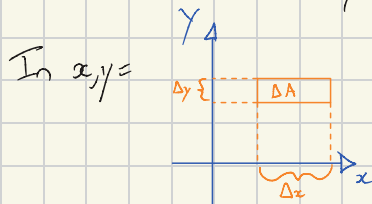
$$\iint_{\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 < 1} dx dy = \iint_{u^2 + v^2 < 1} ab du dv = ab \iint_{u^2 + v^2 < 1} du dv = ab \times \text{area}(\text{unit disk}) = \pi \times ab$$

$\Delta x = \frac{a}{2} \Delta u$ and $\Delta y = \frac{b}{2} \Delta v$ $\Rightarrow du dv = \frac{1}{ab} dx dy \Leftrightarrow du dv \cdot ab = dx dy$

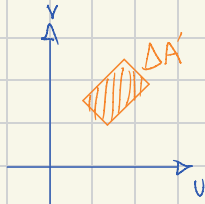
In general: find scaling factor ($dx dy$ vs $du dv$)

Example 2: $u = 3x - 2y$
 $v = x + y$ } we do this sometimes to simplify the integrand or bounds

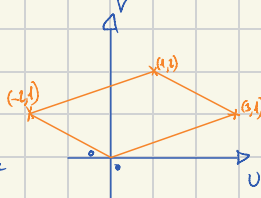
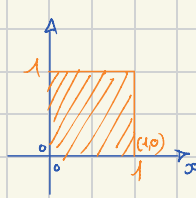
Relation between $dA = dx dy$ and $dA' = du dv$



In u, v :



Area scaling factor here (in this case) does not depend on the choice of the rectangle because we are doing a linear change of variables.



$$A' = \begin{vmatrix} 3 & 1 \\ 1 & 1 \end{vmatrix} = 2$$

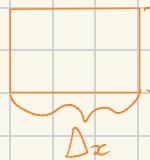
For any other rectangle, area in u, v coordinates is 2 times bigger than in x, y coordinates
So $dA' = 2 dA \Leftrightarrow du dv = 2 dx dy$

$$\iint \dots dx dy = \iint \dots \frac{1}{2} du dv$$

General case: $u = u(x, y)$
 $v = v(x, y)$

$$\begin{bmatrix} \Delta u \\ \Delta v \end{bmatrix} = \begin{bmatrix} u_x & u_y \\ v_x & v_y \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix}$$

Small rectangle in x, y coordinates



$\Delta y \Leftrightarrow$ Small parallelogram in u, v coordinates



Same argument as before, then:

$$\begin{aligned} \langle \Delta x, 0 \rangle &\rightarrow \langle \Delta u, \Delta v \rangle \sim \langle u_x \Delta x, v_x \Delta x \rangle \\ \langle 0, \Delta y \rangle &\rightarrow \langle \Delta u, \Delta v \rangle \sim \langle u_y \Delta y, v_y \Delta y \rangle \end{aligned} \quad \left. \vphantom{\begin{aligned} \langle \Delta x, 0 \rangle \\ \langle 0, \Delta y \rangle \end{aligned}} \right\} \text{area} = \det \begin{bmatrix} u_x & u_y \\ v_x & v_y \end{bmatrix} \Delta x \Delta y$$

Jacobian: $J = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}$ Then $du dv = |J| dx dy = \underbrace{\left| \frac{\partial(u, v)}{\partial(x, y)} \right|}_{\text{absolute value}} dx dy$

Example 1: polar coordinates

$$\begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \end{aligned} \quad \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} x_r & x_\theta \\ y_r & y_\theta \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r \cos^2 \theta + r \sin^2 \theta = r$$

So $dx dy = |r| dr d\theta = r dr d\theta$

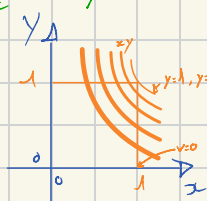
Remark: $\frac{\partial(u, v)}{\partial(x, y)} \cdot \frac{\partial(x, y)}{\partial(u, v)} = 1$ So we can compute the easiest one!
Inverse of this one

Example 2: Compute $\int_0^1 \int_0^1 x^2 y dx dy$ by changing to $u=x$ and $v=xy$

① Find area element (J): $\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} 1 & 0 \\ y & x \end{vmatrix} = x$, so $du dv = x dx dy$ ($x > 0$)

② Integrand in terms of u, v : $x^2 y dx dy = x y \frac{1}{x} du dv = v du dv = v du dv$ (or $du dv$)

③ Bounds: $\int_0^1 \int_v^1 v du dv$
 u changes
 v constant = xy constant



On switch to uv picture

